Sahaj Bhadja

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Summary

Database-focused software engineer with 2+ years building **SQL-backed systems**, query-tuned services, validation pipelines, and developer tooling; strong in PostgreSQL, MySQL, C/C++, Java, Python, Linux, Git, and performance debugging.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Designed **multi-tenant SQL schemas and access patterns** for 500+ users, applying indexing, query tuning, and validation checks to reduce dashboard latency from **1.2s to 350ms**.
- Built **data integration and reconciliation workflows** processing 5,000+ monthly jobs, improving reliability through SQL transformations, error handling, CI/CD releases, and production debugging notes.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed **database-backed APIs and batch workflows** handling 10,000+ daily requests, integrating React, Flask, MySQL, file ingestion, and model outputs into reliable grading services.
- Debugged **query, persistence, and ingestion failures** across production workflows, documenting root causes, test cases, and fixes that reduced instructor feedback turnaround by **60%**.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **Python analysis pipelines** over 100,000+ records, producing reproducible metrics, logs, and quality reports for model evaluation, data validation, and debugging reviews.
- Documented **experimental data flows and edge cases** for researchers, improving reproducibility, reducing handoff ambiguity, and strengthening review quality across iterative technical changes.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **API and ETL microservices** with Python, SQL, AWS Glue, and S3, improving response time by **30%** while supporting scalable data movement.
- Maintained **20+ automated test suites** using Jenkins, PyTest, and Mockito, increasing CI reliability by **68%**, reducing regressions, and improving release confidence.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built **MERN and database-backed modules** with REST APIs, WebSockets, and relational/NoSQL persistence, supporting 2M+ synchronized records across modular workflows.
- Automated **Docker-based build and test flows** with Jenkins, cutting release time by **50%** and improving repeatable deployments, debugging, and documentation.

Skills

Languages: C, C++, Java, Python, SQL, JavaScript, TypeScript, Bash
Databases: PostgreSQL, MySQL, SQL Server, MongoDB, relational schema design, indexing, query tuning
Database Systems: MVCC/coursework familiarity, transactions, replication concepts, query planning basics, storage and WAL concepts
Systems & Tools: Linux/Unix, Git, Docker, CI/CD, Jenkins, GitHub Actions, profiling/benchmarking familiarity
Backend: REST APIs, Flask, FastAPI, Node.js, ETL pipelines, data validation, documentation, production debugging

Projects

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Built an **AI-powered SQL workspace** supporting 10+ databases, accelerating schema exploration, query authoring, and database management through Java services and TypeScript interfaces.

Big-Brain Live | *PostgreSQL, FAISS, LLaMA-2, GCP*

- Built a **PostgreSQL-backed document retrieval platform** over 50K+ records, combining semantic indexing, metadata persistence, and summarization workflows to reduce review time by **60%**.

CPU Scheduling Simulator Live | *C++, React.js, JavaScript*

- Implemented **C++ scheduling algorithms** including FCFS, SJF, SRT, RR, priority, and MLQ with metric calculation and visualization across 1K+ process simulations.